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Escola Técnica Superior de Enxeñaría
Master in Artificial Intelligence

Deliverable 1

Lab I1, I2 & I3

Course:
Trustworthy and Explainable AI (XTAI)

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Overview

This document corresponds to **Assignments I1, I2 and I3** of the course **Explainable and Trustworthy AI** [**P4251104**] [**2025/2026**], within the **Master in Artificial Intelligence**.

The main objective of this work is to study and apply different approaches to **interpretability** and **trustworthiness** in machine learning, combining **post-hoc explanation methods** with **interpretable-by-design models**. In **I1**, we analyse model behaviour through **SHAP (SHapley Additive exPlanations)** on a clinical classification task, producing both **global** and **local** explanations to understand how individual features influence predictions. In **I2**, we explore **Fuzzy Inference Systems (FIS)** generated with **GUAJE**, examining how design choices such as **regular vs induced partitions** and **linguistic simplification** affect the rule base, interpretability, and predictive behaviour. Finally, in **I3**, we design a **human-grounded evaluation** of explanation interpretability using a questionnaire, where a panel of expert participants assesses the clarity, usefulness, and perceived faithfulness of the explanations.

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1 I1 — Introduction to SHAP

1.1 Exercise I1.1

Problem

Try to build some alternate versions of the Decision Trees we obtained, by changing the parameters on the constructor call. You don't need to do this for Regressors, so refer to this documentation. You don't need to be exhaustive: trying two or three is enough. You may keep limiting their depth to keep their interpretation manageable (but you don't need to). Plot these new decision trees and compare them to the two trees (dtc and dtc5) given as illustrative examples in the previous cells. Are they significantly different? Extract some conclusions as to why/why not, and summarize a few key insights onto the diabetes prediction problem you can distill from the trends shown by your set of trees.

Answer

To solve this exercise, we trained two additional Decision Tree classifier variants and compared them against the original models:

- **Entropy-based tree:** we changed the splitting criterion to `criterion="entropy"` while keeping `max_depth=5`, to assess how using Information Gain (instead of Gini) affects the learned split hierarchy.
- **Pruned (min-leaf) tree:** we enforced `min_samples_leaf=20` (with `max_depth=5`) to prevent leaves supported by very few samples and reduce overfitting, yielding more robust decision rules.

We then contrasted these variants with the original unbounded tree (`DecisionTreeClassifier()`) and the depth-limited baseline (`max_depth=5`) by inspecting their text exports, focusing on which variables appear near the root versus those that only emerge deeper in the tree.

The main conclusions drawn from analysing the exported decision-tree tests, where the decision process is shown in a structured way (depending on the value of a variable, the path follows a specific branch) are the following:

1. **Dominance of Glucose:** Across all models, regardless of the splitting criterion (Gini vs. Entropy) or pruning settings, *Plasma glucose concentration* consistently appears as the root node. This reinforces that Glucose is the most discriminative feature for diabetes prediction in this dataset. Nonetheless, the Entropy-based tree selects a lower initial threshold (127.50) than the Gini-based trees (143.50), indicating that information gain optimises the first split differently.
2. **Impact of pruning (`min_samples_leaf`):** The pruned variant is noticeably more interpretable than the original unbounded tree. By requiring each leaf to contain at least 20 samples, the model avoids deep, highly specific branches that are likely driven by noise or overfitting. As a result, the learned rules become more stable and emphasise robust predictors (e.g., BMI and Age) rather than tailoring decisions to outliers.

3. **Metric sensitivity (Gini vs. Entropy):** Although the global structure remains broadly consistent (first Glucose then BMI then Age/Pedigree...), the Entropy-based model tends to introduce slightly different decision boundaries and reorders feature usage at deeper levels. In particular, it appears to prioritise the Diabetes Pedigree Function differently compared to the Gini-based depth-limited baseline (`dtc5`).

We can conclude that the prediction of diabetes in this Pima Indians dataset is driven primarily by metabolic thresholds (Glucose and BMI). The complex branches found in the deeper levels of the unpruned tree likely offer diminishing returns in accuracy while hurting interpretability. The pruned tree (VARIANT 2) offers the best balance, providing clear, consistent rules without getting lost in specific patient details.

1.2 Exercise I1.2

Problem

Now you have an alternative explanation tool against which you can contrast the insight you extracted on Exercise I1.1, and some knowledge onto how the Random Forest is predicting. Try to interpret the graphs corresponding to the Classifier alternatives:

- Do the SHAP values for the Decision Trees match your inspection of their graphical representation?
- Do you find significant differences between the Decision Trees and the Random Forest?

It is a good thing if you read SHAP documentation in order to understand better how to interpret the generated plots. Keep in mind that the answer for any of these questions may be negative, but you should still try to theorise as to why (justify your responses).

Answer

For this exercice we computed SHAP values for the test set using `shap.TreeExplainer` for three classifiers (an unbounded Decision Tree, a depth-limited Decision Tree, and a Random Forest).

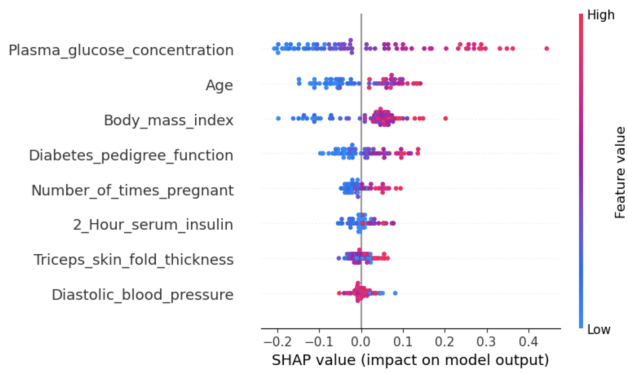


Figure 1: SHAP summary plot for the unbounded Decision Tree.

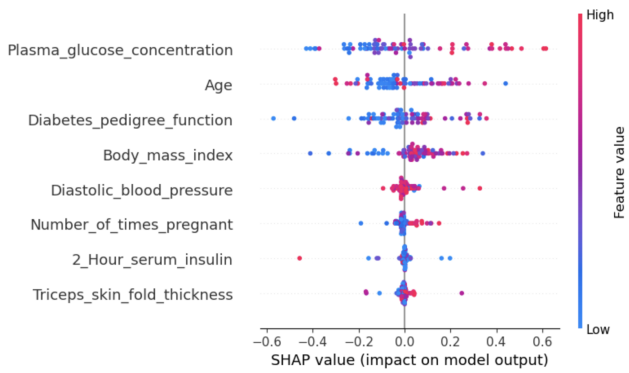


Figure 2: SHAP summary plot for the depth-limited Decision Tree (`max_depth=5`).

When looking at the figures shown, the relationship between the tree structure and the SHAP explanations becomes clear. Recall that **SHAP** (SHapley Additive exPlanations) assigns to each feature an additive contribution that quantifies how much it pushes a prediction up or down with respect to a baseline; in the summary plot, the *distribution* of SHAP values across the test instances shows both overall importance (wider spread and larger magnitudes) and whether a feature tends to increase or decrease the predicted risk depending on its value.

In our case, the Decision Tree's SHAP values are consistent with its representation. By inspecting the tree, the earliest splits are driven by *plasma glucose concentration*, which already suggests that this variable dominates the decision process. This is exactly what we observe in the SHAP summary plot: Glucose exhibits the largest spread and magnitude of SHAP values across samples, indicating the strongest global impact on predictions. Conversely, features that only appear in deeper branches (or that are rarely selected for splits) present much smaller SHAP contributions, since the model has fewer opportunities to use them to change the output. Overall, the features ranked at the top in the SHAP plots match the variables responsible for the first separations in the tree, confirming that the explanation aligns with the model's actual decision logic.

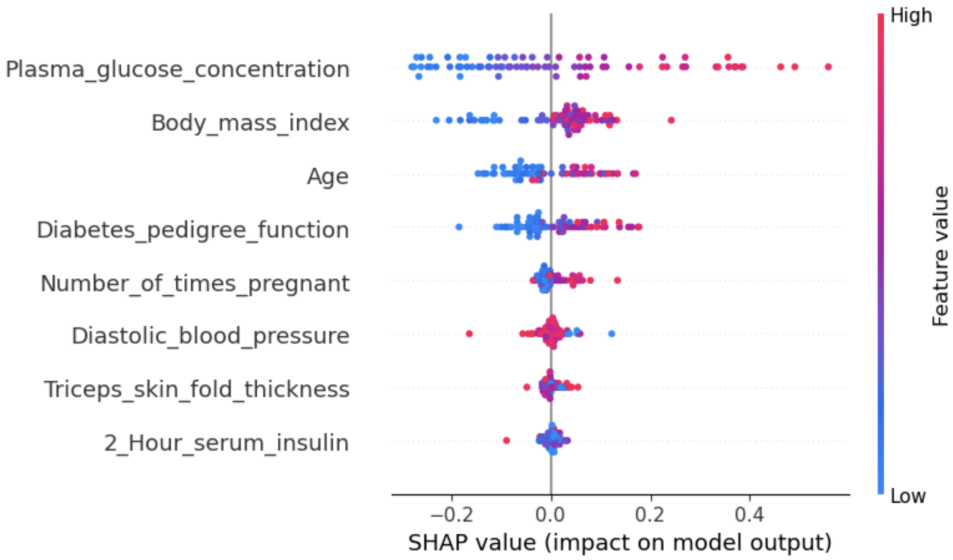


Figure 3: Mean absolute SHAP values on the test set for the Random Forest (global feature importance).

When we focus on the Random Forest (see the SHAP plots above), the differences with respect to the single Decision Trees are more quantitative than qualitative. In both cases, the overall pattern is largely preserved: **glucose remains the most influential feature** by a clear margin, and the next most relevant variables (e.g., BMI, age, and diabetes pedigree function) are essentially the same set that appears near the top of the Decision Tree explanations. What changes is mainly the **relative ordering and magnitude** of these mid-ranked features: depending on the model, some of them may swap positions or show slightly different spreads, but they do not introduce a fundamentally new hierarchy.

This is also visible in the distribution of SHAP values: the least important features remain tightly clustered around 0, indicating that they contribute very little to the final prediction for most test instances. Therefore, based on these figures, the Random Forest does not exhibit a

dramatically broader "importance spread" across many additional variables; instead, it yields an explanation that is consistent with the trees, maintaining a similar feature structure while smoothing or slightly rebalancing the contributions among the same main predictors.

1.3 Exercise I1.3

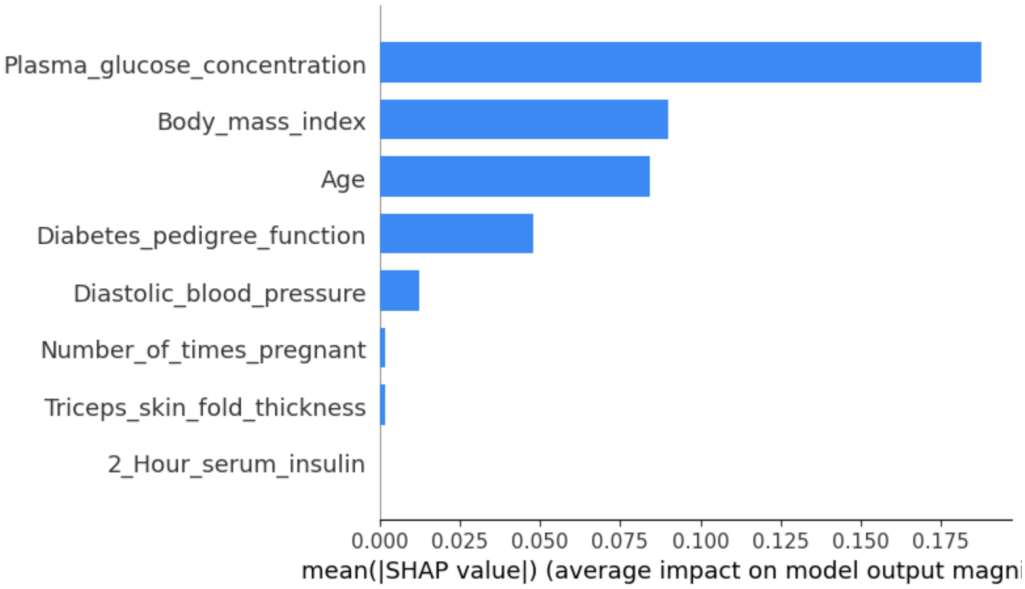
Problem

Have a look at SHAP's API reference. So far, we have only used the basic summary plot, but one of the library's greatest strengths is its rich plotting environment. Try your hand at generating new plots (both global and local) following the code given during the section. Do you think they offer better information than the summaries we were using? Which plotting options do you think would be the best at explaining the problem at hand to a layperson? Would your answer change if the explanations were meant for an expert?

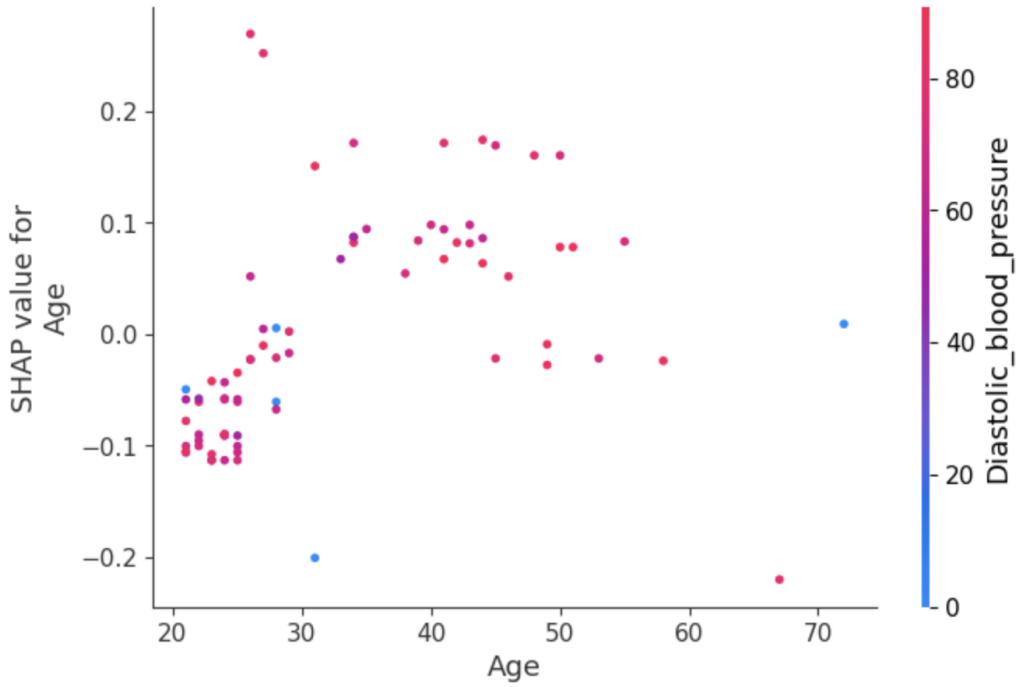
Answer

We generated several complementary SHAP visualizations beyond the beeswarm summary, including a global bar plot for feature ranking, dependence plots for feature effects across value ranges, local waterfall plots for single-instance explanations, and decision plots to compare multiple instances. We then assessed which plots best support different audiences (lay vs. expert) depending on whether the goal is simplicity, behavioural insight, or detailed diagnostics.

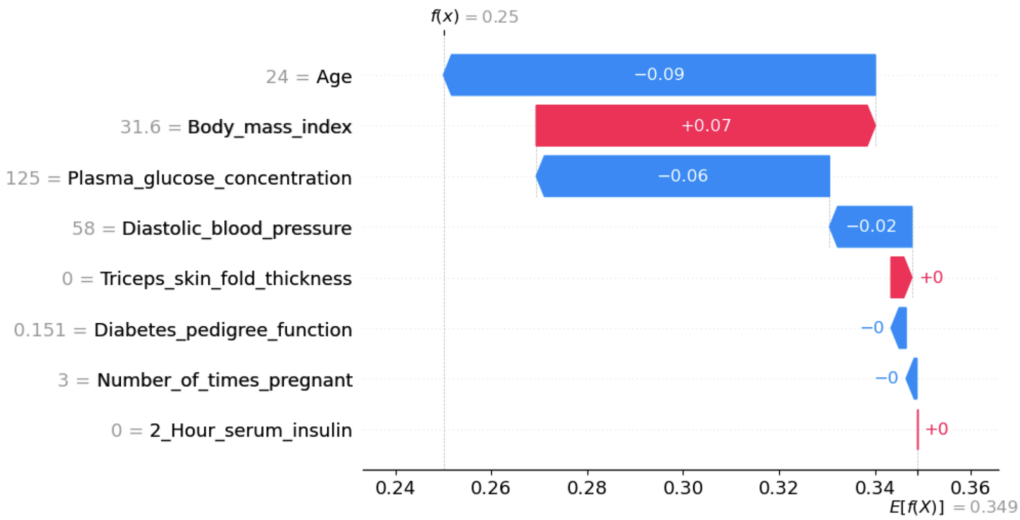
This additional SHAP plots provide more actionable information than the basic summary (beeswarm) alone, because they answer different questions. The bar summary gives a clean global ranking of the most influential features.



On the other hand dependence plots reveal how a feature's effect changes across its value range (and can expose non-linearities or interactions).



For local reasoning, waterfall (or force) plots are especially informative because they show how individual feature contributions accumulate from the base value to the final prediction, making it clear why the model predicted a specific outcome.



After understanding the different information each representation provides we can anwer that for a layperson, we would prioritize the bar summary for a simple "top factors" explanation and a waterfall plot for one representative case, since both are easy to narrate ("glucose pushes the prediction up, BMI adds more, age slightly reduces it," etc.). For an expert audience, we would switch to the beeswarm summary plus dependence plots to inspect distributional effects and conditional behavior, and add decision/waterfall plots across multiple instances to compare patterns (e.g., true positives vs. false positives) and better diagnose model behaviour.

1.4 Exercise I1.4

Problem

Now that you've reached the end of this first practical session, you have all the tools needed to autonomously perform the following task. Use the code below to import the SONGS dataset, present on your workspace, which contains 2017 data instances representing songs from a Kaggle competition. The dataset is intended to serve for the classification problem of predicting if the song will be 'Like' or 'Dislike'. With the dataset imported, follow the process laid out through this notebook to explain the global behaviour of the model. Then, find a song you like (and another song that you don't like) on the dataset (using the code provided), and locally explain the predictions made by a decision tree with a good interpretability-accuracy trade-off and Random Forest. Discuss all your findings.

Answer

To solve this exercice we imported the SONGS dataset and trained two classifiers: a Decision Tree selected via a depth sweep guided by an interpretability–accuracy trade-off (using the number of leaves as a proxy for complexity) and a Random Forest as a stronger black-box baseline. Then we computed global SHAP explanations (bar and beeswarm summaries, plus dependence plots for top features) and produced local explanations (waterfall) for two selected song instances, comparing how each model distributes feature contributions.

After applying SHAP to the PIMA dataset, the same explainability workflow is transferred to the SONGS dataset.

The model selection for the Decision Tree is guided by an interpretability– accuracy trade-off: as depth increases, cross-validated performance improves only marginally after a certain point, while complexity (measured by the number of leaves) grows quickly. The Pareto-style comparison makes this "diminishing returns" effect visible: beyond a moderate depth, the model becomes harder to read without gaining much predictive value, so a shallow tree is preferred as the final interpretable classifier.

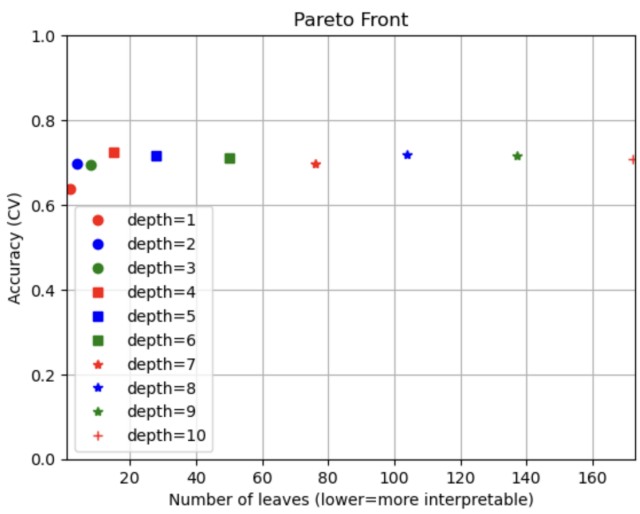
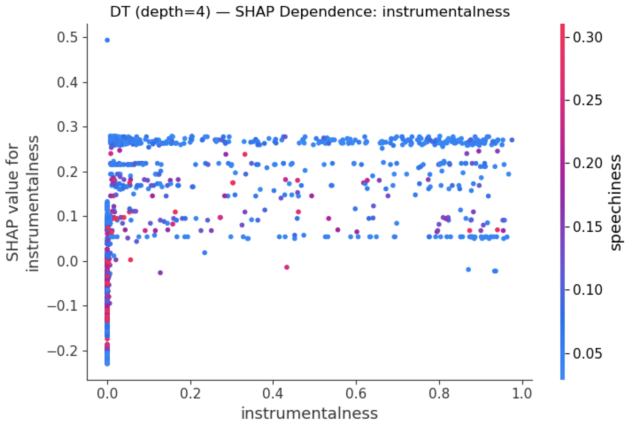
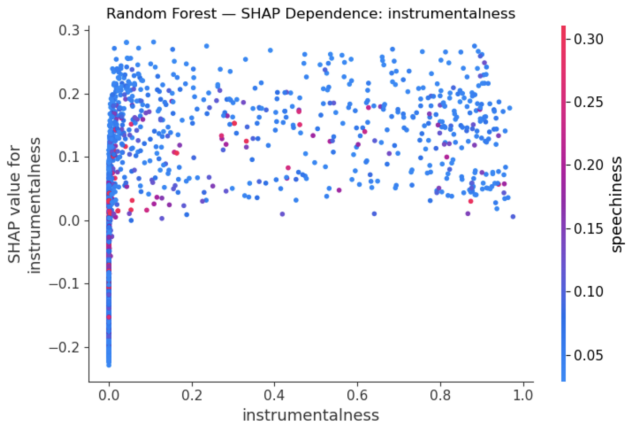


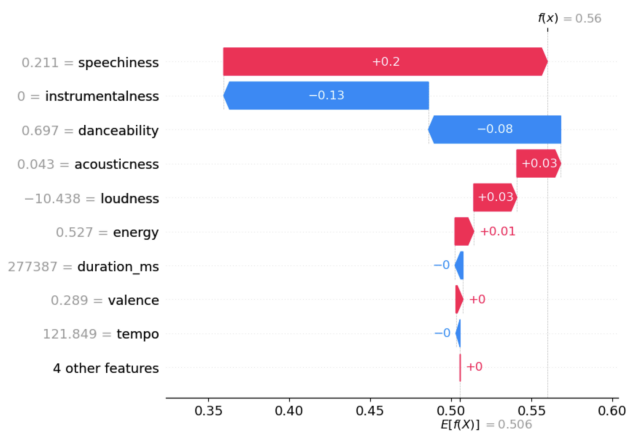
Figure 4: Pareto-style comparison

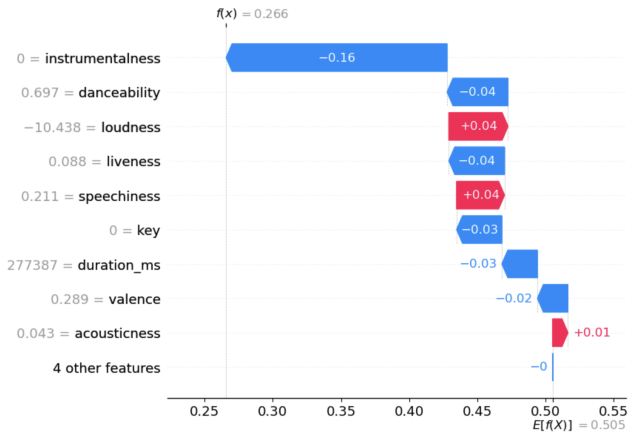
The global SHAP plots (bar and beeswarm) provide the main behavioural summary. The bar plot ranks features by average absolute impact, making clear which audio properties drive the Like predictions. The beeswarm adds directionality and variability: it shows whether high or low values typically push the model towards Like and how consistent that effect is across songs. In this dataset, instrumentalness appears as the strongest discriminator, with loudness, speechiness, danceability, and duration_ms also contributing substantially. Compared to the Decision Tree, the Random Forest tends to distribute importance more broadly, which is expected because the ensemble aggregates many trees trained on different samples and feature subsets, allowing it to exploit weaker but complementary signals that a single greedy tree might underuse.





Dependence plots refine these conclusions by revealing non-linear patterns and interactions. Instrumentalness often behaves like a threshold feature: songs concentrated near zero show mixed contributions, but higher instrumentalness values tend to consistently increase the SHAP contribution towards Like. Speechiness typically exhibits a clearer monotonic tendency, where larger values increasingly support Like. Loudness requires careful interpretation because it is measured in negative dB; its effect can reflect stylistic clusters (quieter vs louder production) and may interact with features such as energy. Local explanations (waterfall plots) then connect these global trends to concrete song choices: the Decision Tree usually yields shorter, easier-to communicate explanations but can be sensitive near the decision boundary, while the Random Forest spreads credit across more features, often producing more stable decisions and higher confidence when several musical cues align.





2 I2 — Building Explanations in Natural Language (Fuzzy Systems)

2.1 Exercise I2.1

Problem

Observe the system we ended up with:

- It seems that we "lost" two fuzzy variables, considering that IRIS has 4 features. This is obviously an effect of simplification, but why did simplification take those two out?
- Consider the data distributions you see on the histograms for Petal Length and Petal Width. Then, consider the fuzzy partitions shown for those two variables. What do you think the system did to simplify the partitions for Length? Do you think that whatever it is, it is justified in light of the underlying data distribution?

Tip 1: Take a look at the rule base! Tip 2: Take a look at data distribution and potential correlations!

Answer

By inspecting the baseline Knowledge Base alongside the generated data distributions (pair plots) and correlation matrix, we can draw the following conclusions regarding the simplification process:

- **Feature Selection (Lost Variables):** The variables Sepal Length and Sepal Width were not inherently "lost", but rather strategically pruned by the algorithm because they were uninformative. As depicted in the correlation heatmap (see Figure 5), Petal features exhibit a significantly stronger correlation with the target class than Sepal features. The model learned that examining the Petals provides sufficient discriminative power to identify the Iris species. Discarding the redundant Sepal features effectively reduces the complexity of the Rule Base, yielding a final model that is vastly more interpretable and computationally efficient without sacrificing accuracy.
- **Partition Simplification:** The system intelligently merged linguistic terms based on the natural clustering of the data. As observed in the pair plot (see Figure 6), the "Setosa" species (Class 1) forms an isolated cluster characterized by very low values. The Rule Base demonstrates that Petal Width alone can cleanly isolate this cluster (*RULE 1: IF Petal Width IS low THEN Class IS 1.0*). Consequently, Petal Length does not require a distinct "Low" partition to separate this class; it only needs to distinguish between the overlapping "Average" and "High" clusters to separate Versicolor and Virginica (*RULE 2: IF Petal Length IS more or less (average) AND Petal Width IS average THEN Class IS 2.0*). Therefore, merging the "Low" values into the "Average" term for Petal Width is entirely justified, as it streamlines the partitions while perfectly adapting to the underlying data distribution.

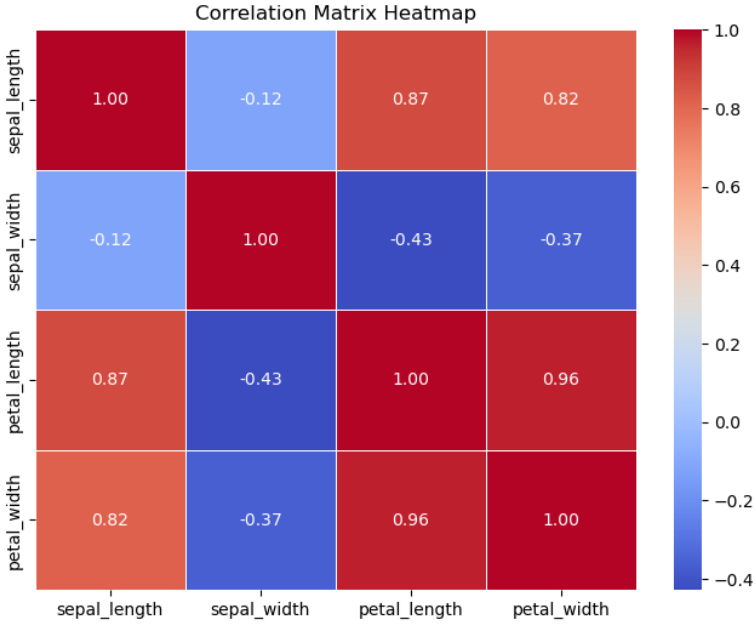


Figure 5: Correlation Matrix Heatmap of the Iris dataset features.

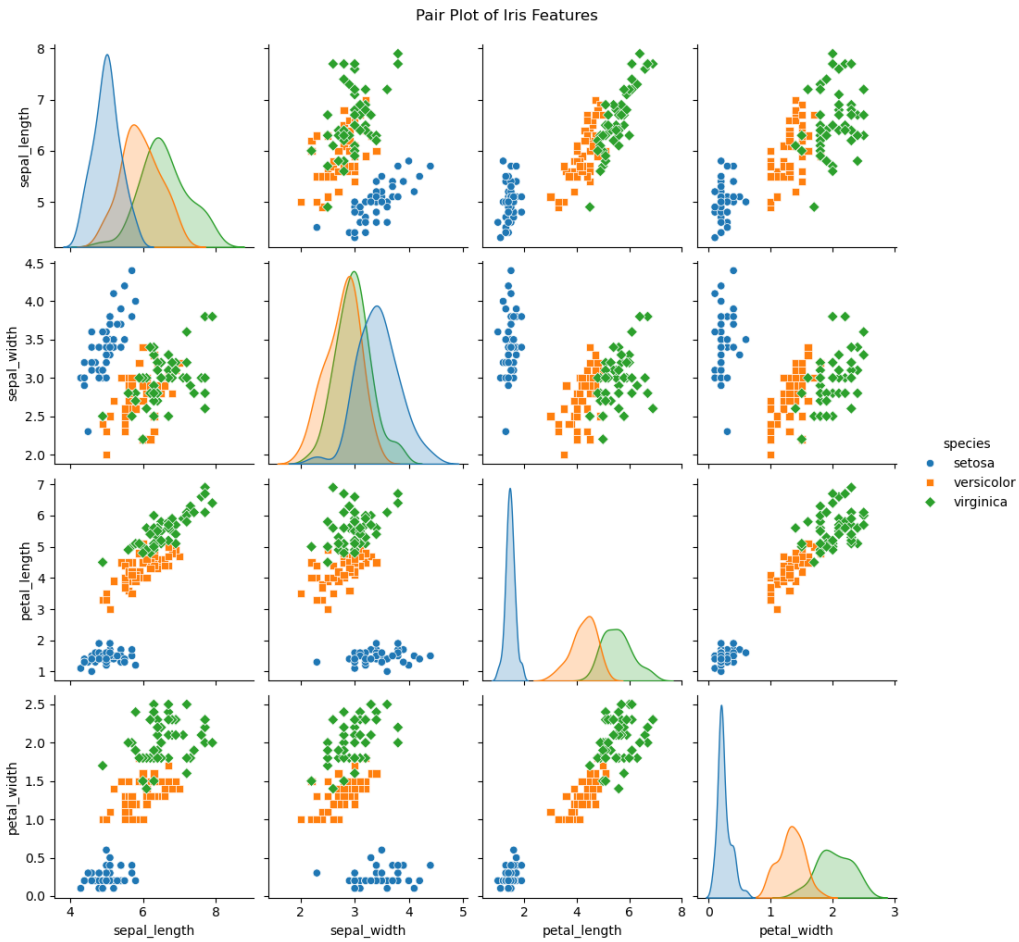


Figure 6: Pair plot illustrating feature distributions and class separation.

2.2 Exercise I2.2

Problem

Compare the features of these two new systems against the one we took as baseline (RP_FDTP_S):

- What differences appear in each case? Why?
- If you had to choose only one alternative, without further information, which would you choose? Why?

Answer

By evaluating the Knowledge Bases and Rule Bases of the two alternative systems (RP_FDTP and SP_FDTP_S) against the baseline (RP_FDTP_S), we observed distinct differences stemming from the simplification and partitioning strategies applied:

- **Differences in System Features:**

  - **RP_FDTP (Without Simplification):** This system exhibits a noticeably more complex Rule Base. It contains an additional rule overall and involves more antecedent comparisons within individual rules. This highlights the role of the simplification step present in our baseline: simplification acts as a post-processing algorithm that identifies redundancy, effectively merging "sibling" rules to make the system simpler and more interpretable without losing granularity or accuracy.
  - **SP_FDTP_S (Induced/Scatter Partitions):** The most significant change here lies in the shapes and placements of the membership functions. In the baseline RP (Regular Partitions) system, the triangles and trapezoids are arbitrarily and evenly spaced, simply dividing the variable's domain into equal parts regardless of how the data is actually distributed. Conversely, the SP system generates irregular partitions that dynamically align with the clusters (the "mountains") visible in the data histograms. Consequently, linguistic terms like "Low", "Average", and "High" in the SP system accurately represent real clusters of data points, as these induced partitions learn their shapes directly from the dataset (typically using clustering algorithms or quantiles).

- **System Selection:** If forced to choose only one alternative without further information, we would select the second system (**SP_FDTP_S**). This system represents the optimal balance for Explainable AI: its scatter partitions faithfully model the true, underlying statistical nature of the dataset, making its linguistic terms much more realistic, while the included simplification step ensures that the final set of rules remains compact, readable, and highly interpretable for end-users.

2.3 Exercise I2.3

Problem

Taking the code above as an example, replicate the inference using the alternative systems defined above (no simplification / induced partitions). Observe and discuss the results obtained. Which of the two modifications created a higher deviation from the baseline? You may predict for several instances if it helps you draw conclusions.

Answer

To evaluate the impact of the different system architectures, we replicated the inference process for a subset of 16 Iris instances across three models: the baseline (RP-FDTP-S), the no-simplification variant (RP-FDTP), and the induced-partitions variant (SP-FDTP-S). By analyzing the predicted classes and raw output values, we drew the following conclusions regarding system deviations:

- **Impact of Simplification (RP-FDTP vs. Baseline):** Removing linguistic simplification yielded exactly the same numerical predictions as the baseline for all tested instances. This confirms that the simplification step primarily optimizes the structural readability and efficiency of the Rule Base by removing redundancies (e.g., merging sibling rules), without fundamentally altering the decision-making behavior or the accuracy of the model on this sample.

- **Impact of Partitioning Strategy (SP-FDTP-S vs. Baseline):** In contrast, switching from regular to induced (scatter) partitions introduced clear deviations from the baseline. While the majority of instances remained correctly classified, certain borderline cases flipped their predicted class. For example, instances 70 and 149 were classified as *Versicolor* under the baseline but correctly shifted to *Virginica* under the induced partitions.

Therefore, the partitioning strategy (regular vs. induced) produces a significantly higher deviation from the baseline than disabling simplification. This is logically sound, as induced partitions modify the very shapes and boundaries of the membership functions to fit the real data distribution, which directly alters the firing strengths of the rules that determine the final classification.

To further validate this, we inspected the rule activation for a representative borderline case (Instance 145). In both the baseline and the no-simplification models, three identical rules fired with the exact same strengths (e.g., a rule for Class 3 dominating with a weight of 0.833). However, in the induced-partitions model, the adapted membership functions completely changed the inference landscape: only two rules fired, both strongly supporting Class 3 (with weights of 0.449 and 0.550). This demonstrates how tailoring the partitions to the data clusters forces the system to rely on a different, often more accurate, set of active rules for borderline instances.

2.4 Exercise I2.4

Problem

Check the Pareto Front above:

- Why do you think the FIS may be placing above the DT in the tradeoff?
- Induced Partitions seems to be slightly better than Regular Partitions, at least in light of the results shown. Can you imagine a situation in which, with this same Pareto Front, the Regular Partitions alternative would be preferred? Why?

Answer

To rigorously analyze the accuracy-interpretability trade-offs, we evaluated three fuzzy inference systems against three tree-based machine learning baselines using cross-validation. By plotting an interpretability proxy (number of fuzzy rules versus the number of decision tree leaves) against classification accuracy, we constructed a Pareto Front. Based on this comparative evaluation, we draw the following conclusions regarding the model placements:

- **FIS vs. Decision Trees (DT):** A Fuzzy Inference System (FIS) positions itself favorably above the Decision Tree in the tradeoff curve because it employs soft boundaries. A single data instance can partially match multiple linguistic terms and activate several rules simultaneously. This characteristic allows the FIS to approximate smooth, non-axis-aligned decision boundaries using a very compact set of general rules (especially after undergoing FDTP pruning and linguistic simplification). In contrast, a Decision Tree relies entirely on crisp, orthogonal threshold splits. To mimic a smooth transition or a diagonal boundary between classes, a tree must make many successive cuts, which

rapidly inflates the number of terminal leaves and severely degrades human interpretability. Consequently, the FIS achieves comparable accuracy with a fraction of the structural complexity.

- **Regular vs. Induced Partitions:** Although the Pareto Front indicates that induced partitions yield slightly better accuracy by dynamically adapting membership functions to the training data, there are practical scenarios where the Regular Partitions alternative would be strictly preferred. Induced partitions are highly tailored to the specific dataset, making them more susceptible to sample bias, noise, or distribution shifts during real-world deployment. In heavily regulated or high-stakes environments, Regular Partitions are advantageous because they enforce stable, uniformly spaced, and semantically consistent linguistic labels (e.g., absolute thirds for "Low", "Average", and "High"). This uniformity makes the model inherently more robust against data drift, easier to communicate to non-technical stakeholders, and simpler to validate and audit. Therefore, one might consciously choose Regular Partitions to guarantee interpretability consistency over a marginal, potentially fragile gain in accuracy.

2.5 Exercise I2.5

Problem

You have, just above, a complete, comprehensive example on how to define, import and predict/explain for a given problem. Now, using the GUAJE online tool, for which your teachers should have given you access instructions (see the beginning of this notebook), create your own FIS for the PIMA dataset you employed on the previous practical session (you don't need to download it, Online GUAJE already has it pre-loaded). Integrate it into your workspace, and then repeat the task above with it to generate some explanations. Would you prefer this alternative to the SHAP/Visual approaches we used on the previous practical? Why?

Answer

To comprehensively evaluate our approach, we first generated a Fuzzy Inference System (FIS) for the PIMA dataset using GUAJE Online with a 5-fold cross-validation strategy. We integrated the five exported JFML (XML) models into our notebook and ran inference over the full dataset to compute aggregated confusion matrices and classification metrics.

Cross-Validation Performance: Across the five folds, overall accuracy remained stable within a narrow band (0.736–0.775), suggesting that the learned rule bases are robust to training splits. However, a clear class-wise pattern emerged: the systems are consistently better at recognizing the "Healthy" class (recall 0.87–0.94) than the "Diabetes" class (recall 0.41–0.57). The aggregated confusion matrix confirms this conservative tendency (TN=2262, FP=238 vs. FN=682, TP=658). The model tends to avoid false positives at the cost of producing false negatives, frequently labeling borderline cases as healthy. While providing a solid, interpretable baseline with high specificity, screening applications would likely require reducing these false negatives (e.g., by adjusting the decision threshold or biasing training toward the positive class).

FIS vs. SHAP (Explainability Comparison): Regarding whether we prefer this FIS alternative to the SHAP/visual approaches from the previous practical, we view it not as a strict replacement, but as a choice driven by application requirements. We would prefer

the FIS when absolute interpretability and auditability are the primary priorities. Because the model is transparent by design, explanations follow directly from the rule base, providing actionable counterfactuals. Conversely, we would prefer SHAP when predictive performance is paramount, allowing us to deploy a highly accurate black-box model while still obtaining local/global feature attributions. Given that our current FIS produces many false negatives, high-stakes medical screening might dictate using a more complex model alongside SHAP, whereas transparent decision policies would favor the FIS approach.

Global and Counterfactual Explanations (Complex Model): To fully explore the capabilities of linguistic generation, we evaluated a secondary, unsimplified complex model in GUAJE to extract factual and counterfactual explanations using the `FCFEXPGEN` library:

1. **Global Interpretation:** Unlike opaque models, printing our Knowledge Base explicitly revealed 13 understandable rules. We can seamlessly verify the medical logic, such as *Rule 1*, which logically dictates that a low plasma glucose concentration strongly points toward a negative diabetes test.

2. **Factual Explanation:** For our test instance (Patient 0), the system correctly predicted a positive diagnosis. The library explicitly stated the driving factors: *"The test instance is of class tested positive because plasma glucose concentration is label 4 and age is old."* This perfectly matches the graphical output, where Rule 10 dominated the inference process with an activation weight of 0.93.

3. **Counterfactual Explanation:** To understand the decision boundaries, the system generated an alternative scenario: *"The test instance would be of class tested negative if body mass index were high and age were young."* This highlights Age as a critical risk factor for this specific model, suggesting that if this exact patient had been significantly younger, the risk assessment would have dropped below the positive threshold.

General Conclusion: Throughout this evaluation, we culminated our work by successfully building, evaluating, and interpreting a Fuzzy Inference System for a real-world medical problem. By combining performance metrics (K-fold Cross-Validation) with powerful visual and linguistic XAI tools (GUAJE and FCFEXPGEN), we demonstrated that it is entirely possible to achieve accurate medical inferences while retaining complete transparency, allowing domain experts to fully trust and verify the AI's reasoning.

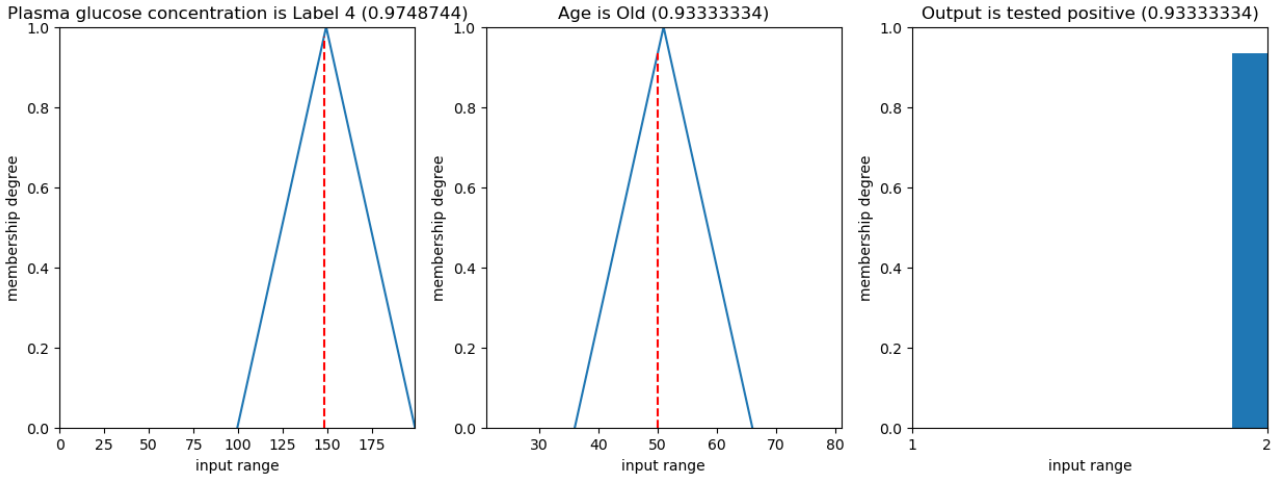


Figure 7: Example of a fuzzy rule application.

3 I3 – Questionnaire: Interpretability of SHAP Explanations

The questionnaire was implemented in Microsoft Forms and can be accessed here: Interpretability of SHAP explanations.

3.1 Objective of the questionnaire

The goal of this questionnaire is to evaluate whether SHAP visual explanations are interpretable and useful for understanding a model trained on clinical variables from the PIMA Indians Diabetes dataset. In particular, the study focuses on two complementary explanation levels: (i) **global explanations**, where participants must interpret what a SHAP summary (beeswarm) plot says about the model's overall behaviour across many patients, and (ii) **local explanations**, where participants must interpret a SHAP waterfall plot for one specific patient and connect it to the model's predicted output.

Beyond checking basic comprehension of the visual encodings (axes, colours, and signs), we also measure perceived clarity, confidence, and perceived faithfulness (i.e., whether users believe the explanation reflects the model's actual reasoning).

3.2 Type of evaluation, assessment, and target population

Type of evaluation

Our study is primarily an **intrinsic evaluation**, because it assesses the explanations themselves (i.e., whether the SHAP plots are understandable, whether users can correctly decode what the axes/colors/bars mean, and whether they can extract the intended message from the explanation). Participants are not asked to perform a full downstream clinical decision-making task with real consequences, nor are we measuring improvements in end-task performance. That said, a small component of the questionnaire can be seen as light extrinsic evidence, since

some items resemble explanation-grounded micro-tasks (e.g., identifying which feature most increased the prediction for the specific patient, or applying a 0.5 threshold to decide the class). These tasks remain tightly bound to the explanation, so the evaluation is still best characterised as predominantly intrinsic.

Regarding whether our evaluation is quantitative or qualitative, it follows a **mixed-methods** approach. On the **quantitative** side, we include multiple-choice comprehension questions that can be scored for correctness and summarised through accuracy rates, as well as rating-scale items (Likert-type) that can be analysed via distributions and central tendency. On the **qualitative** side, we include an optional open-ended question to capture nuanced feedback that fixed options may miss (e.g., requests to clarify whether the SHAP output is shown in probability space or log-odds, or suggestions about how to present the predicted class). Combining both perspectives is important because an explanation can be objectively interpretable (high correctness) yet still be perceived as unclear or untrustworthy by users, and vice versa.

Assessment construct

Following the taxonomy proposed in the assignment, our questionnaire is best described as a **human-grounded assessment**. Participants act as expert evaluators and judge whether the explanations are understandable, useful, and faithful, without being embedded in a real operational pipeline (e.g., actual clinical decision-making with physicians and real patients). This choice is appropriate because the study goal is to evaluate the interpretability of SHAP visualisations themselves, and the target population (XAI-trained users) can reliably assess the intended semantics of the plots.

Target population

The target participants are **expert evaluators in XAI** (students with advanced background in AI and explainability). This choice is appropriate because the questionnaire requires reading technical plots (SHAP summary / waterfall) and answering concept-driven questions. In the collected responses, the population is consistent with an expert cohort: all respondents reported a Master's level education, and English proficiency is mostly B2–C2.

3.3 Choosing the samples to evaluate

The questionnaire includes two types of visual material:

- **Global sample**: a SHAP summary (beeswarm) plot computed over a set of patient instances, used to communicate global feature importance and the typical direction of impact for low vs. high feature values.

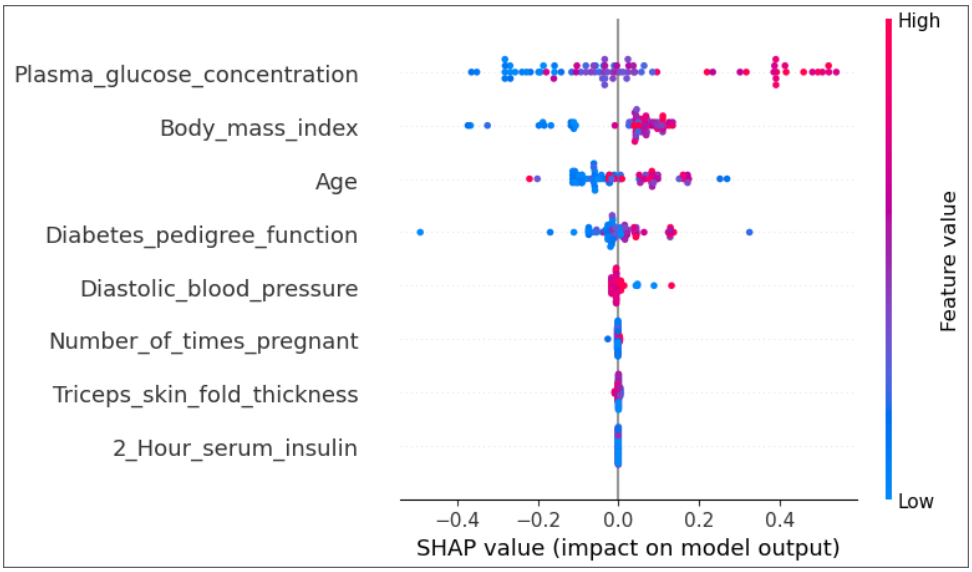


Figure 8: Global explanation shown to participants: SHAP summary (beeswarm) plot.

- **Local sample**: a SHAP waterfall plot for a single patient instance, where contributions from individual features are added to the baseline $E[f(X)]$ to obtain the final model output $f(x)$. The local case was selected to include both positive and negative contributions, so participants must interpret opposing effects rather than a trivial one-sided explanation.

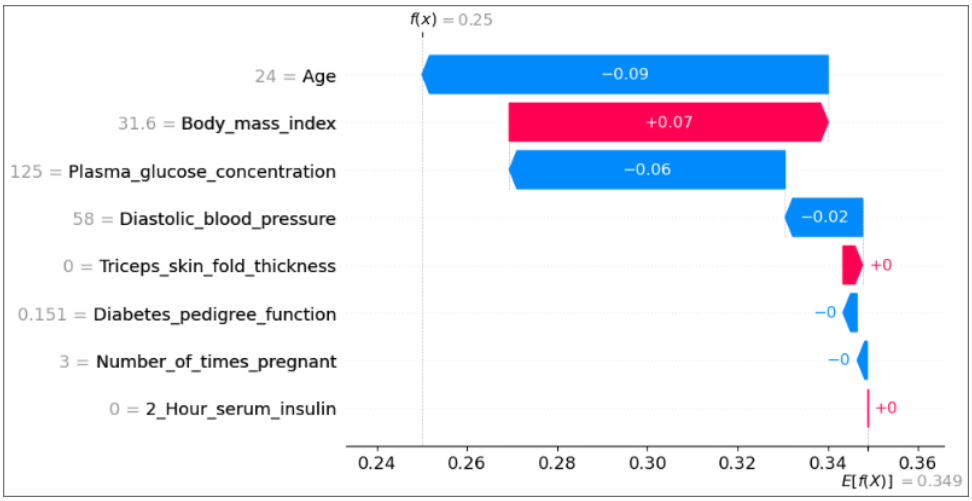


Figure 9: Local explanation shown to participants: SHAP waterfall plot for one patient instance.

The intention is that the global plot tests "how the model behaves in general", while the local plot tests "why the model predicted this for this specific patient".

3.4 Research Questions and Hypotheses

We define the following research questions (RQ) and hypotheses (H), focusing on interpretability and perceived trustworthiness of SHAP explanations:

RQ1: Can expert users correctly interpret the semantics of a SHAP summary (beeswarm) plot and extract correct global conclusions?

H1: Experts will achieve high accuracy on global comprehension questions (at least 80% correct), indicating that the beeswarm plot is interpretable for the target population.

RQ2: Can expert users correctly interpret a SHAP waterfall plot for a single instance and connect it to the model output?

H2: Experts will achieve high accuracy on local comprehension questions (at least 80% correct), including baseline interpretation and identification of the strongest contributors.

RQ3: Do experts perceive the SHAP explanations as clear and as reflecting the model's actual reasoning?

H3: A majority of experts will report positive subjective ratings (clarity, confidence, and faithfulness), with responses concentrated on the top two categories of each scale.

These RQs map directly to the dual nature of explanation evaluation: correctness-based interpretability (RQ1–RQ2) and perceived trustworthiness/usability (RQ3).

3.5 Evaluation Plan

The evaluation plan follows a simple, reproducible pipeline:

1. **Recruitment**: invite participants from an expert cohort (XAI-trained students) and share the questionnaire link. Participation is voluntary and anonymous.

2. **Consent step**: participants must explicitly confirm consent before proceeding (binary yes/no gate).

3. **Global block**: show the global beeswarm plot and ask comprehension questions that require decoding axes, colours, and global feature importance ordering.

4. **Local block**: show the local waterfall plot and ask comprehension questions requiring reading the baseline, identifying feature values shown in the plot, interpreting sign and magnitude, and relating contributions to the final output.

5. **Subjective block**: collect Likert-scale ratings on clarity, confidence, and faithfulness.

6. **Qualitative block**: optional open comment on confusing aspects or suggested improvements.

7. **Demographics**: collect basic background information to contextualise results (education, English level, age range, gender).

Analysis strategy. Quantitatively, we compute per-question accuracy (objective interpretability) and aggregate accuracy separately for global and local blocks. For subjective ratings, we report distributions and check whether they support H3. Qualitatively, we summarise recurring themes in open comments.

Success criteria. We consider H1 and H2 supported if the global/local accuracy surpasses the 80% threshold; H3 is supported if the majority of responses fall into the top categories ("Agree/Strongly agree", "Clear/Very clear", etc.).

3.6 Data Management Plan

The data management strategy follows GDPR-oriented principles (data minimisation, purpose limitation, and limited retention):

- **Data collected**: questionnaire answers and coarse demographics. No clinical or sensitive personal data is collected from participants; only their perceptions and answers about the provided SHAP plots.
- **Identifiers**: the questionnaire is designed to avoid collecting direct identifiers.
- **Storage**: responses are stored in a local course workspace accessible only to the project authors, and used exclusively for this academic assignment.
- **Retention**: data is retained only for the duration needed to write and grade the assignment, after which it will be deleted.
- **Sharing**: results are reported in aggregated form (counts/percentages); no raw identifying information is disclosed.

3.7 Ethics Committee approval process

Although the study is minimal risk (survey about interpretability of plots), it still involves human participants. The institutional process is documented through:

- the **request form** for ethics evaluation ("Formulario de solicitude de informe para estudos con persoas"), which requires describing the study aims, participant recruitment, confidentiality measures, and data handling;
- the **consent clause template** ("Modelo de cláusula de consentimento..."), which provides the required GDPR wording and participant rights (withdrawal, access, deletion where applicable).

In a formal submission, we would prepare a small ethics dossier including: (i) the completed request form, (ii) the participant information sheet and consent text, (iii) the questionnaire content, and (iv) a brief data management plan. The request form itself provides the committee contact channel and highlights key considerations when students are recruited as participants (voluntariness, privacy, and avoidance of undue influence in teacher–student relationships).

3.8 Questionnaire implementation

Why these question types?

We deliberately combined **four-option multiple-choice** items with **Likert scales**:

- Multiple-choice items force a concrete interpretation and allow accuracy scoring. Including plausible distractors (e.g., confusing SHAP value with raw value, or colour with class label) helps detect typical misunderstandings.

- Likert items capture subjective impressions that are important for "trustworthy XAI": a plot can be technically interpretable yet still feel unclear or untrustworthy.
- The open-ended question allows participants to express nuances not captured by fixed options (e.g., requests to clarify whether $f(x)$ is probability or log-odds).

Good practices applied

Several design choices were included to improve measurement quality:

- **Locality of questions**: most questions refer directly to what is visible in the provided plot (axes, colours, a concrete feature value).
- **Balanced distractors**: each multiple-choice item includes distractors that reflect realistic misconceptions rather than obviously wrong answers.
- **Inclusion of "cannot be determined"**: at least one question includes a "not possible to determine" option, discouraging guessing and testing awareness of what SHAP plots do (and do not) contain.
- **Progressive difficulty**: the questionnaire starts with basic encoding (axes/colour), then moves to higher-level reasoning (baseline, additive structure, classification threshold).
- **Reducing straight-lining in rating items**: for the Likert-style questions, we deliberately avoided using the exact same response format everywhere. In particular, the most positive option is not consistently placed on the same side, and we alternate adjective scales (e.g., clear vs. useful vs. confident vs. faithful). This reduces mechanical answering patterns ("clicking the same column"), encouraging participants to read each item carefully and improving the reliability of the subjective measures.

All questionnaire questions and rationale

Below we list every question as presented to participants, followed by the reason it was included.

1. **Consent gate:** "By selecting 'Yes', you confirm that you are at least 18 years old, have read the information above, and consent to participate under these conditions."
   Rationale: ensures explicit informed consent and prevents accidental participation without agreement.
2. **Global Q1:** "What does the x-axis (SHAP value) represent?"
   Rationale: verifies that SHAP values are contributions to the model output, not raw feature units.
3. **Global Q2:** "What does it mean when a point is to the right of 0 (positive SHAP value)?"
   Rationale: checks understanding of directionality: positive SHAP pushes toward the positive class.
4. **Global Q3:** "What does the color of each point encode?"
   Rationale: tests whether participants correctly map colour to feature value (high vs. low).

5. **Global Q4:** "In a beeswarm plot, why are features placed higher or lower on the y-axis?"
   Rationale: checks the concept of global importance ordering (mean absolute SHAP magnitude).

6. **Global Q5:** "Which feature appears to have the smallest overall impact?"
   Rationale: tests whether participants can read global impact from the ranking/spread.

7. **Global Q6:** "For Plasma_glucose_concentration, when the feature value is high, what is its typical influence on the prediction?"
   Rationale: tests combined use of colour (high value) and x-position (sign of contribution) for a key clinical feature.

8. **Global Q7:** "If a feature shows a wide spread of SHAP values, what does that imply?"
   Rationale: checks a more subtle concept: the feature's effect can vary across cases due to context or interactions.

9. **Local Q1:** "What is the baseline value shown in the plot?"
   Rationale: verifies understanding of $E[f(X)]$ as the reference point.

10. **Local Q2:** "How many times has the patient of the plot been pregnant?"
    Rationale: checks whether participants can read the raw feature value from the plot labels and recognise what is explicitly shown.

11. **Local Q3:** "In this plot, a red bar means the feature:"
    Rationale: prevents confusion between red as "high feature value" (beeswarm) vs. red as "positive contribution" (waterfall).

12. **Local Q4:** "What does the length of a bar indicate?"
    Rationale: checks magnitude interpretation: longer bars imply stronger contribution for that prediction.

13. **Local Q5:** "Which feature has the largest positive contribution (strongest push toward the positive class) in this plot?"
    Rationale: tests identification of the top positive driver in a concrete local explanation.

14. **Local Q6:** "If you add up all SHAP contributions to the baseline, you obtain:"
    Rationale: tests the additive principle $f(x) = E[f(X)] + \sum_i \phi_i$.

15. **Local Q7:** "Compared to the baseline, the final prediction $f(x)$ is:"
    Rationale: checks reasoning about net effect (overall upward vs. downward shift).

16. **Local Q8:** "Using the standard threshold of 0.5 on a probability output, this instance would be classified as:"
    Rationale: connects the explanation to an explicit decision rule and tests end-to-end interpretation.

17. **Assessment Q1:** "The provided SHAP explanations helped me understand how the model makes its predictions."
    Rationale: subjective measure of perceived interpretability.

18. **Assessment Q2:** "How clear were the SHAP visualisations overall?"
    Rationale: direct clarity/usability rating.
19. **Assessment Q3:** "After seeing the SHAP explanations, how confident are you in the model's prediction?"
    Rationale: measures whether explanations increase confidence in the output.
20. **Assessment Q4:** "To what extent do you think the SHAP explanations reflect the model's actual reasoning?"
    Rationale: perceived faithfulness/credibility.
21. **Open-ended:** "What, if anything, was confusing or unclear about the SHAP explanations?"
    Rationale: collects concrete improvement suggestions and uncovers unanticipated ambiguities.
22. **Demographics: Gender.** *Rationale:* descriptive reporting of sample characteristics.
23. **Demographics: Education.** *Rationale:* validates the intended expert population.
24. **Demographics: English level.** *Rationale:* controls for potential language-related comprehension effects.
25. **Demographics: Age range.** *Rationale:* descriptive reporting and potential cohort effects.

3.9 Results and Discussion

A total of $N = 11$ expert participants completed the questionnaire.

Objective interpretability (RQ1–RQ2)

Across the 15 multiple-choice comprehension questions (global + local), participants achieved very high accuracy. Aggregating all comprehension items, the total accuracy is $156/165 \approx 94.5\%$ correct.

Global block (RQ1)

For the global beeswarm plot questions, accuracy was $74/77 \approx 96.1\%$. Most items reached 10/11 or 11/11 correct, indicating strong understanding of (i) what SHAP values represent, (ii) what a positive SHAP value implies, (iii) how colour maps to feature value, and (iv) why the y-axis is ordered by importance. Importantly, the three errors in this block were not concentrated in a single item: they were distributed across three different questions, meaning that no specific concept emerged as a systematic source of confusion. Moreover, each of these errors was committed by a different participant, so there is no single respondent driving the mistakes. Taken together, this pattern suggests that the global explanation was, in general, correctly understood by the target population, and that the few incorrect responses are more plausibly explained by occasional lapses of attention or momentary misunderstandings rather than by a consistent interpretability issue in the beeswarm plot itself.

Local block (RQ2)

For the local waterfall plot questions, accuracy was 82/88 $\approx$ 93.2%. Participants correctly read the baseline and interpreted the sign of contributions, and they had no difficulty with the additive structure of the plot (baseline plus contributions) or with applying the standard 0.5 threshold rule. However, one question stands out as the main source of errors: *"Which feature has the largest positive contribution (strongest push toward the positive class) in this plot?"* Only 8/11 participants answered correctly, meaning that 27% (3/11) selected an incorrect option (Figure 10). A closer look at the pattern of mistakes suggests that these participants chose the feature with the largest magnitude contribution overall, but that contribution was negative (i.e., it pushed the prediction toward the negative class), not positive.

A plausible explanation is that this item requires simultaneously two visual cues: (i) the direction/sign of the contribution (positive vs. negative, typically encoded by bar colour and orientation) and (ii) the magnitude (bar length). When users focus primarily on the most salient bar length, it is easy to overlook the sign and interpret "largest contribution" as "largest absolute effect" rather than "largest positive push". This is especially likely when the most prominent bar in the plot happens to be negative.

A simple mitigation for this issue would be to make the plot (or the question) more explicit about the sign criterion. For example, the item could explicitly say "largest **positive** (red) contribution" or "largest contributor that **increases** $f(x)$", or the plot could highlight the top positive drivers with an annotation. Alternatively, the questionnaire could include a paired follow-up that separately asks for the largest negative contributor, reducing ambiguity and ensuring that participants actively distinguish direction from magnitude.

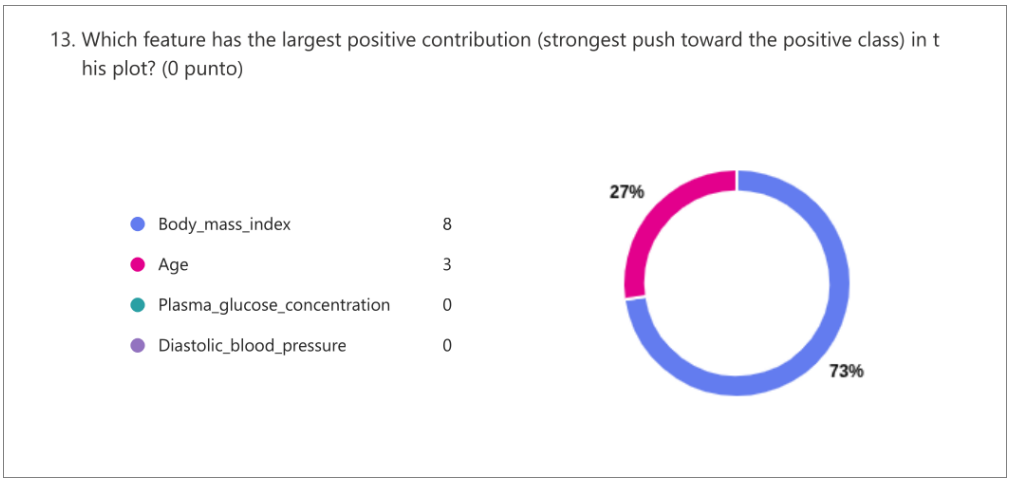


Figure 10: Results of the question 13, with 27% of error.

Conclusion for H1–H2. Both hypotheses are supported: global and local comprehension exceed the 80% criterion by a large margin.

Subjective assessment (RQ3)

Subjective ratings were strongly positive:

- **Helpfulness:** 6 "Strongly agree" and 5 "Agree".

- **Clarity:** 8 “Very clear” and 3 “Clear”.
- **Confidence:** 3 “Very confident”, 5 “Confident”, 3 “Somewhat confident”.
- **Perceived faithfulness:** 7 “Completely”, 3 “Mostly”, 1 “Partially”.

This supports H3, as responses concentrate at the positive end of each scale.

Qualitative feedback

The optional open-ended question received three responses. The main themes were: (i) requesting a brief clarification on whether the model output is in probability space or log-odds, (ii) recommending explicit display of the predicted class next to $f(x)$, and (iii) suggesting adding more local cases (e.g., borderline or misclassified examples) to stress-test interpretability.

Overall discussion

Overall, the results indicate that SHAP beeswarm and waterfall plots are highly interpretable for expert evaluators in this setting. The strongest evidence comes from the convergence between objective performance (high accuracy on comprehension tasks) and subjective ratings (high clarity and faithfulness). The few errors observed point to predictable friction points (e.g., comparing close magnitudes, or potential ambiguity about the numerical scale of $f(x)$), which can be addressed with small presentation changes such as annotating the output scale and explicitly highlighting the top contributing features.